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1. Introduction

1.1 Purpose

InternTrack is a digital platform designed to streamline the SIWES (Student Industrial Work Experience Scheme) process. It enables students, supervisors, and academic staff (including HODs) to complete SIWES activities like logbook documentation and grading fully online. A key security feature is a verification code generated by the SIWES coordinator for authentic student registration.

1.2 Scope

The platform covers the entire SIWES process from:

- Application and verification
- Weekly logbook reporting
- Digital supervisor comments/signatures
- Grading and defense coordination

Key features include:

- Role-specific dashboards (Student, Institution Supervisor, Industry Supervisor, HOD, SIWES Coordinator)
- Smart digital logbook with real-time status tracking
- Email/SMS alerts for pending submissions and grading
- Secure verification during student signup
- Defense result tracking and report generation

1.3 Definitions

Term	Definition
SIWES Coordinator	Approves student placement and generates verification codes
Student	Registers and submits weekly logbooks
Institution Supervisor	Faculty supervisor who monitors and comments on student logs
Industry Supervisor	Company-assigned supervisor during SIWES
HOD	Head of Department, oversees the departmental SIWES process
Logbook	Weekly report submitted by the student
Defense	Oral presentation and final evaluation of SIWES

2. Overall Description

2.1 Product Perspective

InterLink is a web-based platform built using the MERN stack (MongoDB, Express.js, React.js, Node.js). It follows a modular architecture to accommodate multiple users, roles, and real-time workflows.

2.2 User Classes & Characteristics

User	Abilities
Student	Uses verification code to register, fills weekly logbook, views supervisor remarks
Institution Supervisor	Reviews and signs student logs, sends feedback
Industry Supervisor	Optionally comments on weekly performance
HOD	Monitors assigned students and faculty supervisors, validates progress.assigns students to supervisors
Coordinator	Creates verification codes, Upload SIWES Application Letter

2.3 Assumptions & Dependencies

- Users have access to internet-connected devices
- Time synchronization for deadline tracking
- Secure data hosting on cloud infrastructure
- Integration with SMS/Email APIs for alerts

2.4 Project Statement

Paper-based SIWES processes are slow, error-prone, and hard to audit. Stakeholders lack real-time visibility, and students risk lost or incomplete records. **InternTrack** solves this by centralizing data, automating workflows, and enabling secure, multi-role collaboration with clear status tracking and accountability.

2.5 Methodology

- **Process:** Agile/Scrum with 2-week sprints (Design → Build → Test → Review).
- **UX:** User interviews (students/supervisors), low-fi wireframes, clickable prototype, usability tests.
- **Architecture/Stack:** MERN (MongoDB, Express, React, Node), JWT auth, Cloud storage (S3/Cloudinary) for media, CI/CD (GitHub Actions), containerized deploy (Docker) to Vercel/Render + MongoDB Atlas.
- **Quality & Security:** Unit/integration tests (Jest/Supertest), role-based access control, input validation, encryption in transit (TLS), daily backups.
- **Data Protection:** Least-privilege access, audit logs, retention policy, consent and privacy notices.

3. Functional Requirements

3.1 Authentication & Role Management

- Student registration requires a verification code issued by the SIWES coordinator
- Login with email and password
- Role-based dashboard rendering (student, HOD, etc.)

3.2 SIWES Coordinator Dashboard

- Generate verification codes (manually distributed to students)
- Approve/reject student placements
- Upload SIWES Application Letter directly to the student's account after signup

3.3 Student Dashboard

- Register using physically issued verification code

- View SIWES Application Letter uploaded by the SIWES coordinator (private access)
- Fill and submit weekly logbook entries (Week 1–13)
- Upload supporting images (e.g., workplace photos, charts) along with text entries
- View supervisor comments, defense info, and grades
- Track submission status (Submitted / Pending / Needs Review)
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3.4 Institution Supervisor Dashboard

- View student submissions
- Add weekly comments
- Approve or request resubmission of logs

3.5 HOD Dashboard

- See all students under the department
- Track which logs have been signed
- Assign students to faculty supervisors
- Download reports per student or per supervisor

3.6 Industry Supervisor Access (optional)

- May review submitted logs and add remarks

3.7 Grading and Defense

- Add defense schedules
- Assign external assessors
- Input final score after defense
- Track defense history and verdicts

3.8 Notifications

- Email/SMS alert for new logbook, unsent logs, unsigned entries, defense date
- Internal reminders based on deadline calendar

4. Non-Functional Requirements

- SSL-secured connections
- Responsive design (mobile, tablet, desktop)
- Server uptime ≥ 99.5%
- Real-time feedback and live comment saving
- Backup and recovery system
- Complies with Nigerian Data Protection Regulation (NDPR) and GDPR

5. System Models

5.1 Entity Relationship Model (simplified)

- User (id, name, role, email, password, verificationCode)
- Logbook (studentId, weekNo, entry, supervisorComment, status)
- VerificationCode (code, issuedBy, isUsed)
- Assignment (studentId, instSupervisorId, hodId)
- Grading (studentId, score, remarks, date)

6.Literature Review

At Baze University, the SIWES (Student Industrial Work Experience Scheme) process is currently entirely paper-based. Students are issued physical logbooks in which they manually record daily and weekly activities during their industrial attachment. Supervisors then review these paper logbooks during scheduled visits or at the end of the internship period. The Head of Department (HOD) and SIWES coordinator use the same manual records to track student progress and assign grades.

Limitations of the Paper-Based System at Baze University:

- Inefficiency: Manual writing, physical collection, and submission consume time.
- Risk of loss/damage: Paper logbooks can be misplaced, damaged, or incomplete.
- Limited supervision: Institution supervisors may not have real-time access to student work.
- Delayed feedback: Students often wait until a visit or end of SIWES for supervisor comments.
- Data storage challenges: Archiving and retrieving past logbooks is difficult.

Existing Digital Approaches Elsewhere:

- University logbook portals (e.g., in some Nigerian and UK institutions): Provide basic weekly text input and supervisor review, but often lack multimedia support or coordinator-managed verification codes.
- Learning Management Systems (LMS): Platforms like Moodle allow assignment submissions and grading, but they aren't tailored for SIWES processes such as dual-supervisor sign-offs or departmental analytics.
- Field-work documentation apps (engineering, medical, construction): Already use mobile/web tools to capture field logs, often with images and timestamps, but these are not customized for academic internship workflows.

What InternTrack Adds Beyond Current Practices:

- Digitization of Baze's paper-based system with all the same processes but improved efficiency and security.

- Verification code system issued by the SIWES coordinator to ensure only eligible students can register.
- Private upload of application letters directly to student profiles.
- Multimedia logbook entries (images + text) to better reflect student activities.
- Real-time supervisor feedback and digital signatures.
- Department/HOD dashboards with analytics to monitor student compliance and performance.
- Defense grading integration to complete the SIWES workflow digitally.